LOGESH G

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EDUCATION

Bannari Amman Institute of Technology

- Bachelor of Technology in Computer Science and Business Systems **2022 - 2026**
- CGPA: 8.5

EXPERIENCE

Robotics Software Intern, Bengaluru, Karnataka, India **May 2025 - December 2025**

- **MOWITO** - Contributed to a high-precision robotics product by assisting in the development and stabilization of a ROS 2-based software system.

SKILLS

- **Programming Languages:** Python, C, C++, MATLAB
- **Robotics:** Robot Modeling, Kinematics, Motion Planning, Visual Servoing, ROS2 Driver Development
- **Computer vision:** Object Detectors, Object Pose Estimation, Classical Computer Vision
- **Frameworks / Libraries:** ROS2, OpenCV, NumPy, Py-Trees, PyTorch(basics)
- **Tools:** Git, Linux, Docker, Arduino IDE

AREAS OF INTEREST

- Robotic Manipulation
- Computer Vision (Robot Perception)

PROJECTS

- **Visual Servoing using Industrial Arm** *- Robotics and Computer Vision* Dec 2025 - Mar 2026

  Tools used: ROS2, Python, Fanuc TP, and OpenCV
  Description: Developed a visual servoing package for a FANUC LR Mate 200iD/4S industrial manipulator integrating perception and control for object-centric manipulation and servoing. Implemented a real-time EtherNet/IP ROS 2 driver for command streaming and feedback from the FANUC controller. Designed a hybrid IBVS-PBVS visual servoing controller and implemented the task-level architecture using Py-Trees Behavior Trees for modular manipulation workflows.

- **Motion Planning Package** *- Robotics* Mar 2026 - Mar 2026

  Tools used: Python, Pinocchio, NumPy, and MeshCat
  Description: Developed a sampling-based motion planning package for robotic manipulators, implementing RRT and RRT* algorithms from scratch for high-dimensional robot configuration space. Integrated collision checking, nearest-neighbour search, and shortcut-based path smoothing to generate collision-free path, with visualization.

- **SCARA Robotic Arm** *- Robotics and Computer Vision* Oct 2023 - Jan 2024

  Tools used: Python, OpenCV, and PyTorch
  Description: Developed a SCARA robotic arm from scratch with autonomous pick-and-place functionality, integrating YOLOv8 for precise object positioning and a teleoperation system using potentiometers and ROS.

- **Autonomous Robot Manipulation System** *- Robotics and Computer Vision* Mar 2024 - Jul 2024

  Tools used: MATLAB, Virtual Machine, and Gazebo
  Description: Developed a perception and control system for a UR5e robotic arm using MATLAB and Gazebo, integrating YOLOv4 for object detection, pose estimation with depth and point cloud data, and an autonomous control stack for sorting objects into bins.

- **F1TENTH Sim Racing** *- Robotics* Aug 2024 - Oct 2024

  Tools used: ROS2, and Python
  Description: Implemented a LIDAR-based wall-following algorithm to maintain the track center and dockerized the API for seamless integration with the AutoDrive ecosystem and ROS2.

COMPETITIONS

- **RoboCup ARM Challenge 2024 (RoboCup24 - Eindhoven, Netherlands) – Finalist:** Participated in the RoboCup ARM Challenge 2024 (RoboCup24) held in Eindhoven, Netherlands, and was selected as a finalist.
- **1st F1TENTH Sim Racing League (IROS 2024) – Phase 1 Qualified:** Participated and qualified in Phase 1 of the 1st F1TENTH Sim Racing League (IROS 2024), showcasing expertise in autonomous racing algorithms.
- **Flipkart Grid 5.0 - Robotics Challenge – Finalist:** Participated in the Flipkart Grid 5.0 Robotics Challenge, demonstrating innovative solutions in warehouse robotic arms and was selected as a finalist.
- **SAE Aerothon 2023 - Finalist:** Participated in SAE Aerothon 2023, contributed to the computer vision part of the autonomous drone, and was selected as a finalist.

CERTIFICATIONS

- Convolutional Neural Networks – Coursera

LANGUAGES KNOWN

- English - R, W, S
- Tamil - R, W, S